



Realtek Smart Roaming

User Guide

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1. Objective

Roaming in 802.11 is entirely driven by clients' decision and most clients do not process roaming until connection to current associated AP is almost lost.

Realtek smart roaming is to resolve 802.11 roaming sticky problem, refine the overall performance of wireless networks.

2. Method

- Realtek smart roaming is to find the most suitable AP which can provide the best performance for each client.
- Each AP updates its WLAN information including the information of detected clients to the centralized controller (one of APs in the network) periodically.
- The centralized controller evaluates all APs that can detect the same client all the time, and helps this client to roam when current AP is not the most suitable AP for it.

3. Limitation

- The maximum number of APs can be supported is 10.
- The maximum number of associated clients to the network can be supported is 50.
- Only roaming over the same band of different AP devices can be supported.
- Only the SSID and encryption of all test APs can be supported.

4. Configuration and Deployment

4.1. Obtain Service Overlapping

A proper basic service area overlapping between each AP is highly suggested to make sure clients can roam smoothly. Deploy all the APs by making sure the signal strength level of all neighbor APs at the geometric center is **-72dBm**, refer to the figure below.

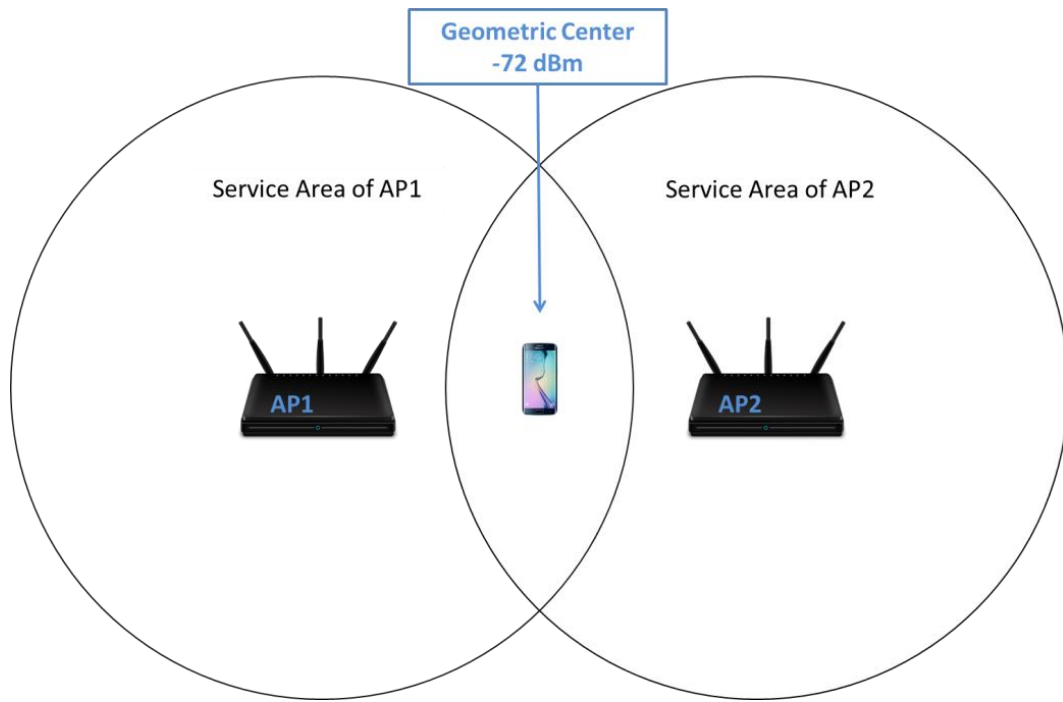


Figure 1 Basic Service Area Overlapping

4.2. DHCP Client of LAN Port

The IP Address of LAN port for each AP needs to be in the same subnet and different from each other. Go to *TCP/IP -> LAN SETTING -> DHCP*, set it to be **DHCP Client**, refer to the figure below.

LAN Access Point	SETUP	WLAN1	WLAN2	TCP/IP
LAN SETTING	LAN Interface Setup			
WAN SETTING	This page is used to configure the parameters for local area network which connects to the LAN. You may change the setting for IP address, subnet mask, DHCP, etc..			
	IP Address: 192.168.1.254			
	Subnet Mask: 255.255.255.0			
	Default Gateway: 0.0.0.0			
	DHCP: Client			

Figure 2 Set LAN Port to be DHCP Client

4.3. Same SSID and Security

Go to *WLAN -> BASIC SETTING -> SSID* and *WLAN -> SECURITY -> Encryption* of every single AP to configure the SSID and Encryption to be the same, refer to figures below.

Realtek	SETUP	WLAN1	WLAN2	TCP/IP
BASIC SETTING	Wireless Basic Settings -wlan1			
ADVANCED	This page is used to configure the parameters for wireless LAN clients which may connect to the wireless network. You can change wireless encryption settings as well as wireless network parameters.			
SECURITY	☐ Disable Wireless LAN Interface			
ACCESS CONTROL	Band: 5 GHz (A+N+AC)			
WDS SETTING	Mode: AP MultipleAP			
SITE SURVEY	Network Type: Infrastructure			
WPS	SSID: Smart_Roaming			
	Channel Width: 80MHz			

Figure 3 Same SSID

WLAN Access Point	SETUP	WLAN1	WLAN2	TCP/IP
BASIC SETTING	Wireless Security Setup -wlan1			
ADVANCED	This page allows you setup the wireless security. Turn on WEP or WPA by using Encryption to protect access to your wireless network.			
SECURITY	Select SSID: Root AP - Smart_Roaming			
ACCESS CONTROL	Encryption: WPA2			
WDS SETTING	Authentication Mode: (RADIUS) Personal (Pre-Shared Key)			
SITE SURVEY	WPA2 Cipher Suite: TKIP AES			
WPS	Management Frame Protection: none capable required			
SCHEDULE	Pre-Shared Key Format: Passphrase			
	Pre-Shared Key:			

Figure 4 Same Security

4.4. Enable Smart Roaming Support

Go to *WLAN -> ADVANCED* of every single AP to enable **Smart Roaming Support** from the WEB-UI, refer to the figure below.

Realtek	SETUP	WLAN1	WLAN2	TCP/IP
Wireless Advanced Settings -wlan1				
These settings are only for more technically advanced users who have a sufficient know should not be changed unless you know what effect the changes will have on your Acco				
BASIC SETTING				
ADVANCED				
SECURITY				
ACCESS CONTROL				
WDS SETTING				
MESH SETTING				
SITE SURVEY				
WPS				
SCHEDULE				
		Fragment Threshold: 2346 (256-2346) RTS Threshold: 2347 (0-2347) Beacon Interval: 100 (20-1024 ms) IAPP: ☒ Enabled ☐ Disabled Protection: ☐ Enabled ☒ Disabled Aggregation: ☒ Enabled ☐ Disabled Short GI: ☒ Enabled ☐ Disabled WLAN Partition: ☐ Enabled ☒ Disabled STBC: ☒ Enabled ☐ Disabled LDPC: ☒ Enabled ☐ Disabled TX Beamforming: ☒ Enabled ☐ Disabled MU MIMO: ☐ Enabled ☒ Disabled Multicast to Unicast: ☒ Enabled ☐ Disabled RF Output Power: ☒ 100% ☐ 70% ☐ 50% ☐ 35% ☐ 15% 802.11k Support: ☒ Enabled ☐ Disabled Fast BSS Transition Support: ☒ Enabled ☐ Disabled 802.11v BSS Transition Support: ☒ Enabled ☐ Disabled Smart Roaming Support: ☒ Enabled ☐ Disabled		

Figure 5 Enable Realtek Smart Roaming Support

Realtek smart roaming can support either roaming over Ethernet (refer to **Section 4.5.**) or roaming over mesh (refer to **Section 4.6.**).

4.5. Roaming over Ethernet

To enable the communication between multiple APs, please make sure the LAN port (only the right-hand-most port on Realtek demo board is WAN port, all the others are LAN ports) of all APs in the network are connected to each other with Ethernet, refer to the figure below.

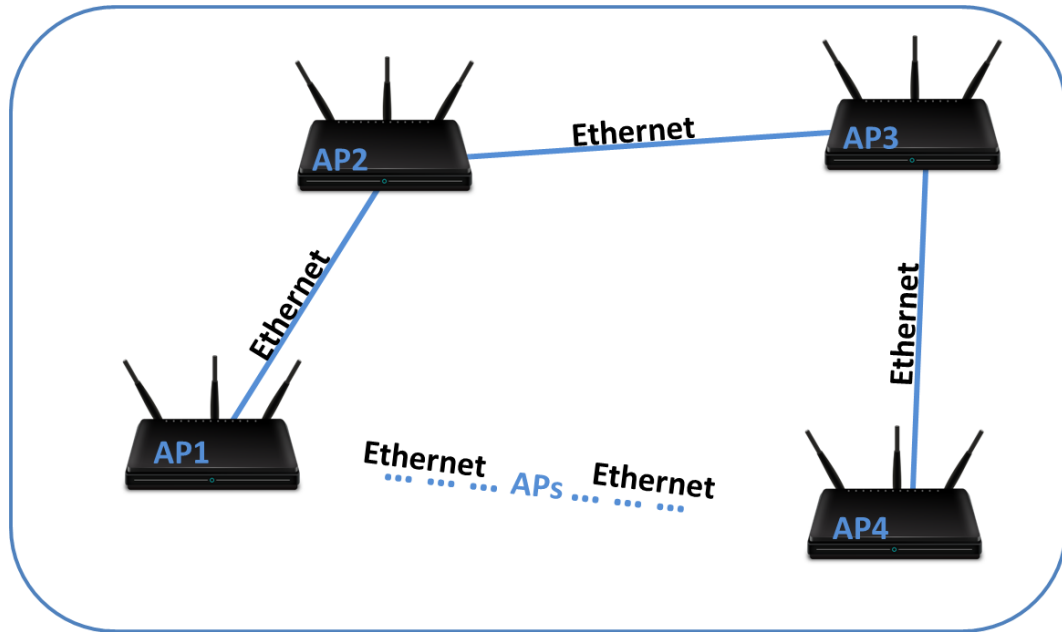


Figure 6 Connect All APs with Ethernet

Connect the LAN port (only the right-hand-most port on Realtek demo board is WAN port, all the others are LAN ports) of one AP in the network to the gateway or DHCP server with Ethernet cable. All the other APs can get the IP address from the same DHCP server through Ethernet connection to this AP.

4.6. Roaming over Mesh

Follow mesh user guide document to configure the channel number, mesh ID, mesh encryption of every single AP to be the same, so that every single AP can communicate with each other through mesh connection.

Connect the LAN port (only the right-hand-most port on Realtek demo board is WAN port, all the others are LAN ports) of one AP in the network to the gateway or DHCP server with Ethernet cable. All the other APs can get the IP address from the same DHCP server through mesh connection to this AP.

5. Roaming Over Mesh Feature Test

5.1. Overview

The centralized controller of Realtek smart roaming system will be auto selected.

To achieve the best overall performance of the network, Realtek smart roaming always helps the client to roam to the most suitable AP which can provide the best throughput for it.

5.2. Test Procedures

1. Setup the mesh network topology to be similar to below **Figure. 11**.

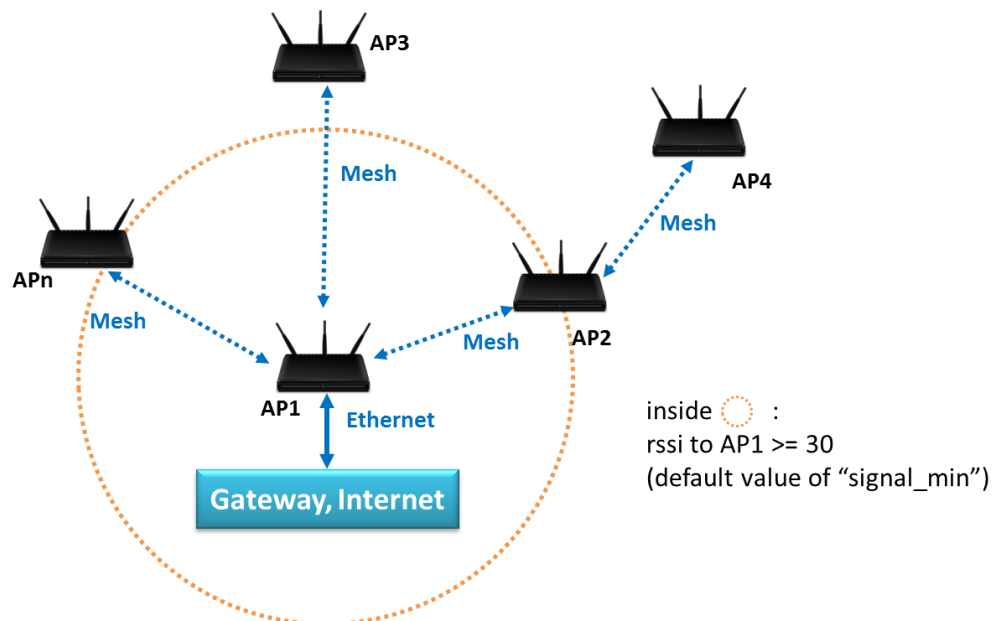


Figure 7 Layout of the Mesh Network

2. Type "**ps | grep capwap**" command from the console of every single AP to check whether it is the centralized controller.
3. Put one wireless client outside , and let it associate to any AP (AP2) in the network.
4. Bring the client to move towards AP1, enter , and stay for a while.

5.3. Pass Criteria

1. There should be only one centralized controller in the network.

- **Normal AP:** *WTP* should be the only result of “ps | grep capwap” check, shown as *Figure. 12*.

```
# ps | grep capwap
1513 root      7712 S      WTP /etc/capwap
1527 root      784 S      grep capwap
#
```

Figure 8 "ps | grep capwap" Result of Normal AP

- **Centralized Controller:** All of *AC*, *AACWTP* and *WTP* should be the result, shown as *Figure. 13*.

```
# ps | grep capwap
1473 root      5824 S      AC /etc/capwap
1476 root      3616 S      AACWTP /etc/capwap
1485 root      7712 S      WTP /etc/capwap
1495 root      784 S      grep capwap
#
```

Figure 9 "ps | grep capwap" Result of Centralized Controller

2. Client should roam from AP2 to AP1 with Realtek smart roaming system.

6. Parameter Adjustment

Most of Realtek smart roaming related parameters are configurable MIBs which can be adjusted during run time by following command:

iwpriv <wlan interface> set_mib <MIB name>=<MIB value>

Example: *iwpriv wlan0 set_mib rssi_threshold=30*

Make the adjustment to take effect by issuing following command from console:

init.sh gw all

Name	Meaning	Type	Default Value
<i>block_aging</i>	duration (in seconds) of AP to execute keeping silent to roaming client	unsigned char	10
<i>signal_min</i>	minimum value of signal strength between the client and the most suitable AP		30
<i>speed_weight</i>	weightage of connection speed when evaluate the AP for client		100
<i>signal_weight</i>	weightage of the signal strength when evaluate the AP for client		1

Table 1 MIB Configuration for Smart Roaming